Running Case (HybridQA): *"What was the original name of the city with an AFL club formed in 1989?"*

Target Ground Truth: "York" • **Modality Fusion:** Table (AFL_Ontario_0) + Wikipedia Bridge (/wiki/Toronto)

Stage 1 (Sec. 3.1): Multi-Source Ingestion

Table T: AFL_Ontario_0

Club	Formed	City
Tor. Eagles	1989	Toronto
Ham. Tigers	1991	Hamilton

→ **Anchor Row 0** (1989)

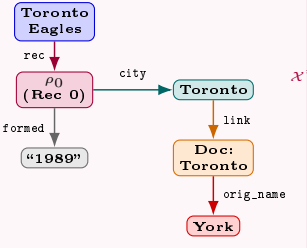
Wikipedia Bridge Expansion:
Toronto → Doc:Toronto

*"...capital of Ontario. Established in 1793 as town of **York**..."*

Retriever Output ($k_1 = 5, k_2 = 3$)
- Coarse candidate rows: $\{\rho_0, \rho_1, \dots\}$
- Cell hyperlinks: /wiki/Toronto
- 100% context preserved (0 doc pruning)

Evidence

Stage 2 (Sec. 3.2): Multi-graph Construction



$\mathcal{X}^*(G_q), p_1$

Star Centroid Hub $\mathcal{G}^*(\rho_i)$ (< 20 ms)
- Centroid binds row cells (Prop. 1)
- Mathematically precludes row conflation
- Deterministic rule text extraction (0 LLM)

Guided Path Scorer $S(p)$ (9 Factors)
- Path p_1: Eagles → ρ_0 → Toronto → York
- Maximal score $S(p_1) = 7.85$ (non-filtering)
- Lossless compression $\mathcal{X}^*(G_q)$ scaffold

Stage 3 (Sec. 3.3): Two-Tier Controlled Engine

Tier 1: L1 Reader (First LLM Call)
- Evaluates compressed multigraph $\mathcal{X}^*(G_q)$ + guided path p_1
- **Controlled Audit Decision:**
  - If determined ($|\mathcal{C}_q| = 1$) → **Fast Exit Answer** (1 turn)
  - If collision ($|\mathcal{C}_q| > 1$) or missing link → **Diagnostic Handoff**

Fast Exit

if collision / ambiguous

Diagnostic Handoff $\mathcal{H}_{\text{diag}}$: {"focus": [$\rho_0, \rho_1$], "check": "town", "type": "collision"}

Targeted Symmetric Bridge Augmentation $\Delta\mathcal{D}$:
Harvests missing candidate passages symmetrically for focus set $\mathcal{C}_{\text{focus}}$

State Capsule $\mathcal{K}_{\text{L2}}$

Tier 2: L2 Reader (Second LLM Call)
- Input: **Minimal Sufficient State Capsule $\mathcal{K}_{\text{L2}}$** (Prop. 2)
- Conclusive factual resolution (eliminates tiebreak guessing)

L2 Verified

Stage 4 (Sec. 3.4): Grounded Provenance

Verified Output Span a^*

"York"
- Exact context-grounded extraction
- Zero ungrounded hallucination

Auditable Provenance $\pi(a^*)$
- Table: AFL_Ontario_0
- Row Hub: ρ_0 (row:0)
- Cell: City: Toronto
- Bridge URL: /wiki/Toronto
- Passage: "...town of York..."

Reverse Grounding Fallback
- Maps span back onto multigraph G_q
- Fallback synthesis if disconnected
- 0.53% empty rate (92.0% faithful traces)